Unstructured Data for Business

Tamara van der Does — Document only to be used in IESE programmes.

Introduction

The first assignment gave you a dataset and a direction. This one starts from a question you care about. The data, the domain, and the framing are yours to choose. The only constraint is that the analysis must go deeper than a single clean spreadsheet allows.

That might mean working with text, time series, or even combining two structured datasets that were never designed to fit together. What matters is that the analytical step you take reflects something you learned on Wednesday that you could not have done on Tuesday.

Your Mission

As a data analyst, your task is to formulate a business question that matters, and answer it using one of the data sources described below or your own data. You may continue with the city you chose in Assignment 1, or pivot to a different domain entirely. The question should reflect your own background and interests: a sector you have worked in, a market you understand, a topic you are curious about.

Going in, write down what you expect to find and why. The most interesting analyses are those where the data confirms, refines, or surprises a worthwhile hypothesis.

Deliverables

On Blackboard, submit a **zip file** named `Unstructured-[FirstName]-[LastName].zip` by **Sunday at 5pm**. It must contain:

1. **A professional memorandum** (2–5 pages) summarizing your findings and recommendations: `Unstructured-[FirstName]-[LastName]-memo.pdf`

2. **A** `README` **text file** covering your business question, where the data came from, and how to reproduce the analysis. You do not need to include the raw data files.

3. **A clean, professional, well-documented Jupyter notebook:** `Unstructured-[FirstName]-[LastName]-nb.ipynb`

In addition, upload your **presentation slides** (`.pdf` or `.pptx`) to Blackboard **before Friday 9:45**: `Unstructured-[FirstName]-[LastName]-slides.pdf`

This document was prepared by Tamara van der Does in June 2026. It is designed to promote class discussion and learning outcomes rather than to illustrate effective or ineffective management of a given situation.

Data Sources

The sources below are available to build upon: use whichever combination best serves your question. You are not required to use them and are welcome to bring in data of your own.

Airbnb Extended Data

The Inside Airbnb project (`https://insideairbnb.com/get-the-data/`) provides two richer datasets alongside the `listings.csv` used in Assignment 1:

- `calendar.csv` — one row per listing per future date, with price and availability. Use it for seasonal pricing patterns, demand forecasting, or lead-time analysis.
- `reviews.csv` — free-text guest reviews with dates. Use it for sentiment analysis, topic modeling, or tracking how perceptions shift over time.

Reddit API

Reddit provides free API access to posts and comments. Collect threads relevant to your chosen domain using the `praw` Python library. Documentation: `https://www.reddit.com/dev/api/`.

> **Note.** Reddit's free-tier rate limits apply. Collect and save your data locally before the submission deadline (do not re-download inside the notebook on every run).

Your Own Dataset

You may use your own dataset, provided it connects to a genuine business question. Good starting points include:

- **Barcelona Open Data:** `https://opendata-ajuntament.barcelona.cat/en`
- **Financial time-series data:** via the `yfinance` Python library (ticker examples: `BTC-USD`, `AAPL`)
- **Kaggle:** `https://www.kaggle.com/datasets` — filter by file type and domain
- **Google Dataset Search:** `https://datasetsearch.research.google.com`
- **World Bank Open Data:** `https://data.worldbank.org`
- **EU Open Data Portal:** `https://data.europa.eu`
- **Network data:** `https://icon.colorado.edu/networks`

If you use your own source, include a brief note in your README explaining where it came from and why it is relevant to your question.

Analysis

Once you have formulated your hypothesis, complete the analysis using the tools covered in Sessions 1–8. Your notebook must include:

1. **Notebook header** (first markdown cell): your hypotheses, to-do list, things tried, and AI used — same format as Assignment 1.

2. **Clear methodology**: the business question, data sources used, key assumptions, and the analytical steps taken.

3. **Data preparation**: types fixed, missing values handled with documented decisions, datasets merged or joined where relevant. Save cleaned files.

4. **Quantitative analysis** that goes beyond Assignment 1: your analysis should draw on at least one technique from Sessions 5–8 (text analysis, data combining, or time series) or demonstrate a more advanced use of tools from the extra notebooks (geographical data and networks) — whichever best serves your question.

5. **Visualizations** that communicate your findings clearly, tied directly to your hypothesis.

6. **Sanity checks** visible in output: sample sizes, date ranges, aggregation totals verified throughout.

7. **Interpretation**: a markdown cell after each major block explaining not just what the output shows, but what it means for the business question.

Creativity and local validity are key. The strongest submissions are those where the hypothesis reflects genuine domain knowledge and the analysis tests it rigorously.

> **Important notes.** Use Claude, ChatGPT, or similar tools freely for debugging and plot styling, but the ideation, hypothesis, and interpretation must be yours. The memo should be substantially your own writing — AI may be used to check grammar and typos, not to draft or restructure it. You SHOULD use the packages covered in class unless you have a strong reason otherwise (justify it in the notebook header). You SHOULD NOT use screenshots for code or text. Log AI use in the notebook header.

Oral Presentation

On Friday morning you will present your analysis to the class in **2 minutes**, using **2 slides maximum**. Think of it as your elevator pitch, the big idea of your executive summary.

Your presentation should cover:

- **The problem:** Question and hypotheses (why we care and what we should expect).
- **The argument:** A plot showcasing one finding (clear and well-designed).
- **The call to action:** What next? (what should the decision-maker do).

Upload **your slides to Blackboard before 9:45 on Friday** as

`Unstructured-[FirstName]-[LastName]-slides.pdf`. You will present from the projected file — no switching between laptops.

> The presentation is assessed separately and counts for **20% of your final grade**. Your written submission (this assignment) counts for 40%. You may update your written submission after the presentations: the deadline for the zip file is **Sunday at 5pm**.

Evaluation Criteria

Written submission — 40 points

Category	Criteria	Points
Technical Analysis *(20 pts)*	Relevance of data chosen for the research question and proper preparation.	4
	Meaningful application of techniques beyond Assignment 1.	4
	Effective visualizations tied to the hypothesis.	4
	Sanity checks and documented decisions throughout.	4
	Interpretation of results and connection to hypothesis.	4
Memo *(20 pts)*	Executive summary: clear problem framing and most important insight.	4
	3–4 data-backed findings in the Findings section.	4
	Actionable recommendations with expected impact.	4
	Implementation plan and call to action.	4
	Structure, clarity, tone.	4
	Total	**40**

Oral presentation — 20 points

Category	Criteria	Points
Content *(9 pts)*	Research question is original and hypotheses are specific.	3
	Finding is specific, data-backed, and clearly stated.	3
	Recommendation is actionable and follows from the finding.	3
Delivery *(11 pts)*	Figure is devoid of clutter and showcases the main point	4
	Narrative has a clear arc and is engaging.	4
	Confident delivery within the 2-minute limit.	3
	Total	**20**

Example Memorandum

TO: *[Recipient / Team]*
FROM: *[Your Name]*
DATE: *[Current Date]*
SUBJECT: *[One sentence stating the core finding]*

Executive Summary

[Briefly introduce the business problem or opportunity you explored, the objective of your analysis, and the most important insight or recommendation. Focus on business value: what matters most to the client or stakeholders.]

Methodology

- **Data sources:** *[List the data used, including any external datasets or APIs.]*
- **Key assumptions:** *[List key assumptions and justify them.]*
- **Approach:** *[Describe your analytical steps (e.g., text sentiment analysis, time series decomposition, dataset merging.)]*

Findings

Insight 1: *[Lead with the finding, then support it with a number. E.g., "Guest reviews mentioning noise increased 34% year-on-year, concentrated in the Eixample district."]*

Insight 2: *[. . .]*

Insight 3: *[. . .]*

Recommendations

Recommendation 1: *[Specific and actionable, tied to your findings. E.g., "Flag listings in noise-complaint hotspots for host coaching."]*

Recommendation 2: *[. . .]*

[Justify each recommendation by tying it back to the findings and explaining the expected business impact.]

Implementation Plan

Resources required: *[Tools, datasets, teams needed.]*

Risk management: *[Data quality, adoption, key caveats.]*

Success metrics: *[E.g., measurable targets for revenue, engagement, or cost reduction.]*

Call to Action

[Conclude with a clear next step for the decision-maker (e.g., approve a pilot, schedule a follow-up, or allocate resources.)]